Canadian Light Source | Centre canadien de rayonnement synchrotron

Linac and LTB Ion Pump Controller Replacement Technical Specification

1.8.33.1 Rev. 0

Date: 2016-May-17

Copyright 2016, Canadian Light Source Inc. This document is the property of Canadian Light Source Inc. (CLSI). No exploitation or transfer of any information contained herein is permitted in the absence of an agreement with CLSI, and neither the document nor any such information may be released without the written consent of CLSI.

Canadian Light Source Inc.
44 Innovation Boulevard
Saskatoon, Saskatchewan
S7N 2V3 Canada

	Signature	Date
	Original on File – Signed by:	
Author	Electrical Engineer	
Reviewer #1	AOD Electrical Engineer	
Reviewer #2	Electrical Engineering Intern	
Reviewer #3	AOD Manager	
Approver	Engineering Manager	

The current version of this document is accessible under the Approved Documents section, on the CLSI Team Site. Employees must verify that any printed or electronically downloaded copies are current by comparing its revision number to that shown in the online version.

TABLE OF CONTENTS

1.0 PURPOSE

The purpose of this technical specification is to outline the technical requirements for the upgrade of the CLS control system for the Linac and LTB ion pump controllers.

The document will be used as a specification for the purchase of the vacuum controller parts required.

2.0 SCOPE

This technical specification will include information about the current control system of ion pump (IOP) controllers and the requirements of the replacement configuration. This will cover the control system requirements, as well as the specifications of the vacuum controller purchase required for the replacements.

The scope of supply for the additional replacement IOP controllers required will be limited to vacuum controllers, rack adapters, and connectors at the controller for high voltage and cables for AC power. Additional components for the control system should be purchased by CLS separately.

3.0 GENERAL BACKGROUND

The Linac and LTB at the Canadian Light Source makes use of ion pumps for maintaining ultra-high vacuum (UHV) pressures (typically less than 1E-8 Torr) within the vacuum chambers. To achieve these pressures, CLS uses ion pumps to pump the vacuum systems when the pressure is at least less than 1E-6 Torr.

The current controllers used to supply power to the ion pumps are quite old and some are >25 years old. The power supplies are failing and the CLS AOD group has requested to replace all of them.

The current configuration of the Linac and LTB IOP controllers is to have 1 to 2 IOPs per channel of the power supply.

ECR911 has been written to describe the changes required. The current power supplies will be replaced entirely, and enough channels will be provided to run all pumps individually rather than using splitters to run high voltage to multiple pumps per channel. However, at this time the high voltage cabling from the controller to the pump will not be upgraded, so the current splitting will remain until a later date. So many of the current channels will be required to supply multiple pumps in the interim.

As this change is fairly disruptive, it should be implemented a zone or section at a time, and preferable coincide with a shutdown of the CLS storage ring. However, maintenance Mondays should also be taken advantage of for this upgrade when the Linac is not in use.

4.0 DEFINITIONS

IOP	Ion Pump
PS	Power Supply
UHV	Ultra High Vacuum
SR	Storage Ring
CLS	Canadian Light Source
IOC	Input Output Controller (RS232 controls computer)
R	Controls Rack (19" in use)
ECR	Engineering Change Request
ECO	Engineering Change Order
Linac	CLS Linear Accelerator
LTB	CLS Linear Accelerator to Booster Ring Transfer Line
AOD	Accelerator Operations and Development
EPICS	Experimental Physics and Industrial Control System

Shall: Indicates a requirement that is to be strictly followed in order to conform to the specification.

Should: Indicates a recommended or preferred requirement. Deviation must be justified.

May: Indicates a possible method or course of action.

5.0 REQUIREMENTS

5.1 CURRENT SYSTEM DETAILS

The Linac and LTB current IOP controller systems use power supplies of the following types (available for exchange/upgrade):

- Quantity: 3 - Perkin Elmer 2220400 (+/- 5.5 KV)
- Quantity: 6 - Constant Voltage 1004498N-9037
- Quantity: 2 - Ultek 60-104L (+ 4.75 KV)

These controllers are currently installed in the Linac and are available for exchange in the SR upgrade, so they are **not** available for the Linac upgrade:

- Quantity: 12 - Varian 929-0080 (- 7 KV)
- Quantity: 1 - Varian 921-0012

Newer temporary replacement controllers currently in use (**not** available for exchange/upgrade):

- Quantity 1: Varian EX929-7009
- Quantity 1: Agilent 929-9401

Some Agilent 4UHV controllers have already been purchased. The following controllers are already available for the replacement project (includes above quantity already in use):

- Quantity of 3 - Agilent 4UHV 4x80W Positive (929-9401)
- Quantity of 1 - Agilent 4UHV 4x80W Negative (929-9400)
- Quantity of 4 - Agilent 4UHV 2x200 Positive (929-9021)

The upgrade shall replace the above controller types with a version that is appropriate for the pumps being supplied. The pump types that are in use in the Linac are:

- Varian VacIon Plus 9191410 (75L/s Diode Positive)
- Varian VacIon Plus 9191510 (150L/s Diode Positive)
- Varian VacIon Plus 9191610 (300L/s Diode Positive)
- Varian VacIon Plus 9191440 (75L/s StarCell Negative)
- Varian REBG-VIP-040-S (40L/s StarCell-StarCell Negative)
- Varian REBG-VIP-075-S (75L/s StarCell-StarCell Negative)
- Perkin Elmer 2070060 (60L/s Diode Positive)

The high voltage connectors in use at this time are a mix of Fischer and SHV 10K type. The interlock connection was not connected through to the pump. In many cases, only a coaxial cable was run from the controller down to the IOP, so these interlock connections are not available. However, in the future the plan would be to replace the IOP high voltage cables running to the IOP, so some of these interlock connections will be available.

At this time, some of the high voltage connectors that connect to the power supplies are SHV10kV. CLS will build adapters from SHV10kV to Fischer connectors to connect to the new controllers, which will have Fischer bulkhead connectors in the controller.

5.2 CONTROLLER REQUIREMENTS

The controller replacements shall provide a current reading for each channel and display the resultant pressure reading down to a pressure of at least 1E-10Torr.

The controller shall be able to be powered with 120V 60Hz AC power. The cord type will fit a NEMA standard 5-15R receptacle.

The controller shall have RS232 communication capabilities. The RS232 communications shall provide pressure information, and vacuum set point status information, and control over all other relevant settings, control parameters and status in the IOP controller.

The controllers shall be capable of at least 5kV voltage of the appropriate polarity for the IOP it is to be used on. Fine variability of the voltage is not a requirement. However, the controller should be capable of fixed voltage settings of approximately 3kV, 5kV and 7kV.

The controller shall support safe connection interlock capabilities. This capability would allow shutting off the high voltage of the power supply when the interlock conductor does not sense connection of the connector at the pump end. This interlock shall by bypass able with an interlock jumper.

The required Fischer high voltage connectors at the controller end will be a quantity of 22 Agilent supplied Fischer connectors (8290705R002). This will cover the parts required for the adapters and future connection to the replacement high voltage cables.

The controllers shall have at least 1 vacuum set point which has at least a normally open dry contact. The contact shall close when the vacuum pressure is less than the set point.

5.3 CONTROLLER SPACE CONTSTRAINTS

There are space constraints within the racks in which the controllers would be used. The available space is summarized in the table below.

The controllers will be upgraded in their current locations as part of this upgrade. So if a controller supplied an IOP in a current location, the replacement would be in the same location. In the future, however, all IOP controllers may be moved into the Power Supply Room location. So the total controller space required should fall under approximately 52U. A 1U space will be required between each row of controllers for ventilation.

Location of Rack	Approximate Rack U's Available for Replacements
R1021.6-06 - Control Room	9U
R0001.1-01 - Zone 1	40U
R0102.1-01 and -02 - Power Supply Room	36U in main rack + 18U in adjacent rack
R0004.1-02 - Zone 4	32U

Table 1: Current Rack Space Available

5.4 CONTROLLER QUANTITIES

The current IOP and proposed Agilent 4UHV controller layout is summarized in the following table. If the "4UHV Available" column says YES, then the IOP controller is already purchased and available. If it says NO, then it will need to be purchased. Depending on the controller purchased, the model number and layout may change.

The IOPs supplied from each current rack will need to remain supplied from that rack. The exception is R0102.1-01 and R0102.1-02 which are next to each other in the power supply room and the racks can be treated as one. Racks R0102.1-01, R0102.1-02, and R0102.1-03 are being replaced as part of this upgrade with two Hammond 36U racks.

The IOC column specifies the existing or new Moxa IOC computer that will communicate to the controllers via RS232.

Rack	IOP	Pump Type	Current IOP Model Number	IOP Size (L/s)	Future PS Equipment Number/Model Number/Channel	4UHV Available	IOC
R1021.6-06 (Control Room)	IOP0001-01	Diode	Varian 9191610	300	PS1021.5-01 (929-9021 CH 1)	YES	IOC1021-504 (existing)
	IOP0001-02	Diode	Varian 9191410	75	PS1021.5-01 (929-9021 CH 2)		
R0001.1-01 (Zone 1)	IOP0001-03	Diode	Varian 9191410	75	PS0001.1-01 (929-9401 CH 1)	YES	IOC0001-101 (new IOC)
	IOP0001-04	Diode	Varian 9191410	75	PS0001.1-01 (929-9401 CH 2)		
	IOP0001-05	Diode	Varian 9191410	75	PS0001.1-01 (929-9401 CH 3)		
	IOP0001-06	Diode	Varian 9191410	75	PS0001.1-01 (929-9401 CH 4)		
	IOP0001-07	Diode	Varian 9191410	75	PS0001.1-02 (929-9401 CH 1)	YES	
	IOP0001-08	Diode	Varian 9191510	150	PS0001.1-02 (929-9401 CH 2)		
	IOP0001-12	Diode	Varian 9191410	75	PS0001.1-02 (929-9401 CH 3)		
R0102.1-02 (PS Room)	IOP0003-01	Diode	Varian 9191410	75	PS0102.1-03 (929-9401 CH 1)	YES	IOC0102-102 (existing)
	IOP0003-02	Starcell	Varian REBG-VIP-40-S	40	PS0102.1-04 (929-9400 CH 2)	NO	
	IOP0003-04	Starcell	Varian REBG-VIP-75-S	75	PS0102.1-04 (929-9400 CH 3)		
	IOP0003-03	Starcell	Varian REBG-VIP-75-S	75	PS0102.1-04 (929-9400 CH 4)		
	IOP0003-05	Starcell	Varian REBG-VIP-40-S	40	PS0102.1-05 (929-9400 CH 1)	NO	
	IOP0003-06	Starcell	Varian REBG-VIP-40-S	40	PS0102.1-05 (929-9400 CH 2)		
	IOP0003-08	Starcell	Varian REBG-VIP-40-S	40	PS0102.1-05 (929-9400 CH 3)		
	IOP0004-01	Starcell	Varian REBG-VIP-40-S	40	PS0102.1-05 (929-9400 CH 4)		
R0102.1-01 (PS Room)	IOP0001-09	Diode	Perkin Elmer 2070060	60	PS0102.1-01 (929-9021 CH 1)	YES	
	IOP0001-10	Diode	Perkin Elmer 2070060	60	PS0102.1-01 (929-9021 CH 2)		
	IOP0003-09	Diode	Perkin Elmer 2070060	60	PS0102.1-02 (929-9021 CH 1)	YES	
	IOP0003-07	Diode	Perkin Elmer 2070060	60	PS0102.1-02 (929-9021 CH 2)		
R0004.1-02 (Zone 4)	IOP0004-02	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-07 (929-9400 CH 1)	YES	IOC0004-102 (new IOC)
	IOP0004-03	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-07 (929-9400 CH 2)		
	IOP0004-04	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-07 (929-9400 CH 3)		
	IOP0004-05	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-07 (929-9400 CH 4)		
	IOP0108-01	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-08 (929-9400 CH 1)	NO	
	IOP0108-02	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-08 (929-9400 CH 2)		
	IOP0108-03	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-08 (929-9400 CH 3)		
	IOP0108-04	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-08 (929-9400 CH 4)		

	IOP0108-05	Starcell	Varian REBG-VIP-75-S	75	PS0004.1-09 (929-9400 CH 1)	NO	
	IOP1300-01	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-10 (929-9400 CH 1)	NO	
	IOP1300-02	Starcell	Varian 9191440	75	PS0004.1-10 (929-9400 CH 2)		
	IOP1300-03	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-10 (929-9400 CH 3)		
	IOP1300-04	Starcell	Varian REBG-VIP-40-S	40	PS0004.1-10 (929-9400 CH 4)		

Table 2: Current IOP and Future Controller Configuration.

5.5 CLS CONTROL SYSTEM REQUIREMENTS

The existing vacuum set-point cables that run to the current controllers to the control system will be re-wired to the new controller set points. Any analog pressures that are currently run to the CLS control system will be removed. The RS232 pressure readings will be used instead.

The IOP Controllers will be remotely communicated to via RS232 with Moxa computers. The CLS convention is to use the Moxa 16 port IOC Computer (DA-662-16-LX). New and existing IOCs will be used for this upgrade. They are summarized in Table 2.

The upgrade will require that a controls driver be developed for communication with the controllers via RS232. The controls driver will provide controller status and control variables via EPICS Process Variables. For the Agilent 4UHV driver, the naming and function of the process variables should support backwards compatibility with the current "Dual" controller process variables. If other models of controller are used, a driver would have to be developed for those as well, if not already available.

The alarms in the CLS alarm handler will be updated to include the vacuum set point status as a warning that the IOP controller may be off or tripped. The vacuum pressures of each IOP are to be added to the data archive. The CLS control screens for SR Injection Status, and individual IOP pump status screens for the SR will be updated.

6.0 WORKPLACE SAFETY AND ENVIRONMENT

The controllers in Zone 1 and 4 will be in the Linac basement at CLS. The control room controllers will be in a cleaner office environment. However, the Linac basement and power supply room will be in a more industrial environment, with water leaks as a possible exposure hazard to the controller.

These controllers will operate at room temperature conditions and are not likely to be exposed to harsh conditions. They will supply ion supply pumps with high voltage cabling lengths of up to approximately 100 meters.

7.0 APPLICABLE CODES, STANDARDS AND PROCEDURES

The controllers will be installed and conform with the standards in the Canadian Electrical Code [1]. The Saskatchewan Interpretations shall be followed, and the required approval marks are outlined in that document. [2].

The electrical design requirements [3] for installation at CLS should be followed, though will not override specifications in this document.

For the controls software and RS232 communications, the standards contained in the Control System Common Specification should be followed, though will not override specifications in this document. [4]

8.0 DESIGN REVIEW, INSPECTION TESTING AND COMMISSIONING

Verification and validation check lists for testing the controllers and the PLC connections shall be developed for the machine protection system controls commissioning. These check lists will be run after the installation of the new power supplies. They will test that the high voltage reaches the IOP with a high voltage meter. They will also test the functionality of the vacuum set points to the PLC and control screens.

The controller vacuum and current readings will be reviewed at time of installation of the new controllers. The vacuum readings may have to be matched to the CCG controller readings if a controller other than the Agilent 4UHV is chosen.

The controls communication via RS232 will be tested to be functioning properly on each pump. This will be part of the commissioning of the controls program. The control screens shall be checked to ensure that the proper IOP status is read back to the correct indicators.

9.0 CALIBRATION

The vacuum pressure readings from the IOP are not required to be calibrated. However, the vacuum pressure readings from the controller should be configured so that they are set to default theoretical presets for the pumps used. The vendor should provide controller settings appropriate for the pumps to which it will be connected.

10.0 RELIABILITY AND MAINTAINABILITY

The vendor shall supply installation and maintenance documentation in English to the CLS for all equipment proposed.

The vendor shall provide a 2 year warranty for the controllers supplied.

11.0 REFERENCES

[1] C22.1-12 Canadian Electrical Code, Part I (2012), Canadian Standards Association

[2] Saskatchewan Interpretations (2012), SaskPower http://www.saskpower.com/wp-content/uploads/2012-Interpretations-FINAL.pdf

[3] Canadian Light Source Electrical Design Criteria – 8.1.16.1

[4] Control System Common Specification – 7.4.39.1

REVISION HISTORY

Revision	Date	Description	Author
A	2016-04-22	Original Draft	Wade Dolton
B	2016-05-10	Adjusted Quantity of Controllers Available	Wade Dolton
0	2016-05-17	Removed Linac Varian Controllers from being available for Linac Upgrade Issued for Use	Wade Dolton